

CONTEXT-AWARE CENTRALIZED FIREWALL (CACF)

Project Development Report

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Abstract

The Context-Aware Centralized Firewall (CACF) is a modern cybersecurity solution that combines traditional rule-based security with machine learning to detect and block malicious activities in real time. By monitoring application-layer requests, it makes adaptive decisions based on contextual data, such as request frequency, user behavior, and data payload patterns. CACF enhances protection against SQL injection, cross-site scripting, and brute-force attacks while offering centralized policy management and analytics dashboards.

1. Introduction

Traditional firewalls rely on static IP and port-based filtering, which often fail against evolving application-layer threats. The Context-Aware Centralized Firewall (CACF) introduces an intelligent and adaptive approach to network security. It operates at the application layer, understanding the semantics of requests and the context of user behavior, providing a unified platform for monitoring, decision-making, and logging security events.

2. Objectives

- Develop a centralized security manager to control firewall operations.
- Integrate a rule-based and ML-based detection mechanism.
- Provide a user-friendly dashboard for real-time analytics.
- Enable dynamic policy management across applications.
- Enhance accuracy and reduce false positives in threat detection.

3. System Architecture

The CACF architecture consists of three major components: 1) Agent (Firewall Layer), 2) Central Manager (Backend API), and 3) Interactive Dashboard (Frontend). The agent intercepts requests and sends them to the manager, where rule evaluation and machine learning detection occur. The manager logs data into a database and provides REST APIs for visualization and control through the frontend interface.

4. Modules Description

Module	Description
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Agent	Intercepts HTTP requests, extracts features, and sends them to the central manager.
Central Manager	Implements FastAPI backend with SQLite database and policy management.
ML Detector	Analyzes requests using a trained model to generate anomaly scores.
Frontend Dashboard	Displays analytics, logs, and policy management through interactive charts and tables.

5. Key Features

- Centralized control and monitoring.
- Hybrid rule-based + ML detection system.
- Real-time analytics dashboard.
- Dynamic security policy creation and management.
- RESTful APIs for communication between components.

6. Technology Stack

- Backend: FastAPI (Python)
- Frontend: React + Tailwind (Lovable.dev / v0.dev)
- Database: SQLite
- ML Module: Scikit-learn, NumPy
- Visualization: Recharts, shadcn/ui
- Deployment: Docker

7. Implementation

The backend provides REST APIs for managing policies, evaluating requests, and logging events. The ML model calculates anomaly scores based on request features. The frontend, built on React and Tailwind, interacts with these APIs to display charts, tables, and dashboards. The system supports CRUD operations for policies and simulates request evaluation via the /check endpoint.

8. Results and Discussion

The CACF system successfully detected and blocked suspicious activities like SQL injections and XSS attacks. The dashboard visualizations allowed easy monitoring of real-time requests and attack trends. Machine learning improved accuracy by dynamically learning from event patterns. The system achieved high responsiveness and scalability during tests.

9. Future Scope

Future improvements include integrating advanced deep learning models for anomaly detection, expanding multi-cloud deployment support, and adding automated alerting systems for administrators. Real-time streaming analytics and adaptive policy recommendations can further enhance security automation.

10. Conclusion

The Context-Aware Centralized Firewall (CACF) demonstrates a powerful blend of traditional rule-based and modern ML-based security methods. It enables intelligent decision-making, central visibility, and adaptive response to threats, marking a significant step toward automated, context-aware cybersecurity systems.